

Sub-Saharan Africa Research and Action Agenda

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Executive summary

Climate change is increasingly recognised as a threat to mental health, compounding risks for poor mental health outcomes and destabilising the conditions needed for good mental health. While research at the intersection of climate change and mental health has proliferated in recent years, the field remains disconnected, uneven and siloed, slowing urgent progress to address the mental health impacts of climate change.

Connecting Climate Minds (CCM) is a Wellcome-funded initiative which aims to cultivate a collaborative, transdisciplinary climate change and mental health field with a clear and aligned vision. Over the last year, we have convened experts across disciplines, sectors and countries to develop regional and global research and action agendas. These agendas set out 1) research priorities to understand and address the needs of people experiencing the mental health burden of the climate crisis, and 2) priorities to enable this research and translate evidence into action in policy and practice. This report presents the research and action agenda for climate change and mental health in sub-Saharan Africa (SSA).

SSA is ecologically diverse with several ecosystems that are highly vulnerable, including communities that are at particular risk to the harmful and threatening effects of climate change. Serious climatic events can result in psychological distress, including traumatic stress, which can develop into chronic psychopathologies and psychiatric patterns over time.¹

Within the context of SSA, there is insufficient research on mental health challenges linked to climate change. There is a need for robust, complex, adaptive, integrated research concerning climate change impacts on mental health, healthcare systems, and policy and practice, in order to implement and strengthen targeted interventions.¹ Throughout the CCM initiative, a total of 31 priority research themes emerged for the SSA region covering myriad areas, as summarised below.

- The magnitude of, and mechanisms driving, climate-related mental health challenges.
- Measuring climate-related mental healthcare seeking behaviour and outcomes.
- The effects of climate-related mental health stigma on interventions and access to care.
- The compounding effects of wars, violence, trauma and epidemics on climate-related mental health challenges.
- Understanding lessons from Indigenous communities on climate change adaptation and mitigation methods that can positively affect mental health.
- Assessing the possibilities for and efficacy of applying medicines, counselling techniques and practices from Indigenous communities to address climate-related mental health challenges.
- Integrating and strengthening climate change and mental health policies, training and awareness initiatives.
- Using new models and methods (e.g., telemedicine, AI, community-based providers) to expand and improve access to treatment for climate-related mental health challenges.
- Evaluating the role and relationship between international agencies, country specific government agencies, academia, employer groups, unions and NGOs in effectively supporting climate change and mental health interventions.
- Assessing the readiness of the mental health workforce and healthcare system to address climate-related mental health challenges.
- The role of, and relationship between, international agencies, country specific government agencies, academia, employer groups, unions and NGOs in effectively supporting climate change and mental health interventions.

- The readiness of the mental health workforce and healthcare system to address climate-related mental health challenges.

Tackling climate change impacts on mental health in sub-Saharan Africa necessitates an integrated approach which synthesises mental health considerations into climate change adaptation and mitigation strategies and vice versa. Multilevel interventions are needed, including: developing and implementing climate policies in sub-Saharan Africa which prioritise mental health; providing support services for affected individuals and communities; and fostering community resilience to better cope with the mental health challenges associated with a changing climate.

Introduction

Context

Climate change and mental health are two of our greatest global challenges, and awareness of the intersection between mental health and the climate crisis has grown rapidly in recent years.² Climate change exacerbates mental health challenges by increasing exposure to extreme heat and the traumas of extreme weather events³, destabilising the conditions needed for good mental health and wellbeing (for example, water and food insecurity, forced migration, polluted air, loss of treasured environments)³, disrupting access to healthcare⁴, and increasing psychological distress through awareness of climate threats and insufficient climate action.⁵ People living with mental health challenges are also particularly vulnerable to the stressors of the climate crisis, such as increased risk of physical heat stress and death during heatwaves.^{6, 7, 8}

In response to the mounting mental health toll of the climate crisis, research in the climate and mental health field has grown rapidly. Nevertheless, key evidence gaps exist in many regions, including the mental health burden attributable to climate change, the pathways and mechanisms underlying these impacts, the co-benefits of climate action for mental health and the best interventions or solutions to support mental health in a changing climate. Climate change and mental health research remains frustratingly disconnected across disciplines, sectors, and geographies, and is unevenly focused on certain topics and global regions.⁹ Moreover, siloed decision making slows the translation of evidence to aligned action across climate and mental health policy and practice.^{10, 11} A more inclusive, connected agenda is urgently needed to generate the evidence to truly understand, monitor and respond to the interconnections between climate change and mental health.

Connecting Climate Minds

Connecting Climate Minds (CCM) is a Wellcome-funded project launched in 2023 to develop an inclusive agenda for research and action in climate change and mental health. The project has two key, intertwined aims. The first is to develop an aligned and inclusive agenda for research and action that is grounded in the needs of those with lived experience of mental health challenges in the context of climate change, to guide the field over the coming years. The second is to kickstart the development of connected communities of practice for climate change and mental health in seven global regions (designated by the Sustainable Development Goals), equipped to enact this agenda. We aim to combine the strengths of a global perspective and regional focus, and bring together diverse disciplinary perspectives into a shared vision that can ensure research is effective at addressing priority evidence gaps and informing changes in policy and practice at the intersection of climate change and mental health.

Through bringing together experts across diverse disciplines, sectors and countries, the CCM team has facilitated the development of a lived experience-informed research and action agenda for the climate change and mental health field in sub-Saharan Africa (SSA).

Objectives of the research and action agenda

The research and action agenda is designed to focus future efforts to help those who are experiencing, or will experience, the compounding mental health challenges of climate change. It aims to support those

who are already responding to these challenges – through communities, research, policy and practice – by building a more connected and collaborative climate change and mental health field. It also aims to empower experts across disciplines and sectors to join and make progress in this area by identifying clear priorities and fostering a more inclusive and transdisciplinary field.

The agenda addresses these aims through three core objectives, which are to:

1. Identify priorities for research that can inform action to meet the needs of people experiencing and responding to the mental health impacts of climate change in sub-Saharan Africa.
2. Identify what is needed to appropriately conduct research and translate evidence to action in policy and practice in sub-Saharan Africa.
3. Build understanding among researchers, funders, and policy experts across disciplines and sectors of their role in furthering climate change and mental health research, and equip them with these clear and actionable priorities.

The regional agenda will be integrated with six other regional agendas to inform a **global research and action agenda** for climate change and mental health. This will ensure that global research efforts and investment in climate change and mental health are grounded in regional-level priorities. Importantly, the global agenda will also integrate insights from agendas developed with and for some of the most affected groups globally, namely Indigenous Peoples, youth, and small farmers and fisher peoples.

Use of the terms climate change and mental health

Climate change, mental health and their intersection are complex and wide-ranging fields. For the purpose of this agenda, we define the scope of these terms as follows.

By **mental health challenges**, we mean thoughts, feelings and behaviours that affect a person's ability to function in one or more areas of life and often involve significant levels of psychological distress. This includes, but is not limited to, anxiety, depression, post-traumatic stress, psychosis, suicidal thoughts and substance misuse.

By **experiences of the effects of climate change**, we mean: 1) experiencing direct impacts of climate hazards, such as more frequent and intense heatwaves, wildfires/bushfires, drought, floods or storms (for example, typhoons, hurricanes, cyclones), and 2) experiencing disruption to the social and environmental determinants of good mental health, such as being forced to move home, not being able to access food or water, losing livelihood or homelands, or disruption to cultural practices as a result of climate change.

Mental health challenges in the context of climate change include:

- How climate change may lead to worsening pre-existing mental health challenges.
- How climate change may contribute to the prevalence or impact of existing mental health challenges.
- How climate change may impact treatment access or effectiveness for those with mental health challenges.
- How climate change may lead to new mental health challenges.

The convening work of CCM presents a key opportunity to build our understanding of diverse perspectives, framings and terminologies in SSA, which we have sought to reflect within the research and action agenda.

Background to Connecting Climate Minds

Regional Community of Practice

In SSA, CCM is led by a Regional Community Team (RCT) responsible for convening diverse expertise across the region and building regional capacity to create and enact the research and action agenda. The structure of the RCT is outlined below.

Regional Community Convenor (RCC)

Purpose: Responsible for developing and delivering project activities in the region, including convening and supporting a regional community of diverse expertise.

Member: Prof Kenneth Anchang Yongabi, School of Health Sciences, Imo State University & Claretian University, Nigeria.

Co-convenors

Purpose: Bringing additional breadth of expertise across disciplines and countries (i.e. organisations spanning climate expertise, stress neuroscience and mental health expertise, in different sectors), providing technical advice and review, and supporting project delivery.

Members: Prof Monika dos Santos, Department of Psychology, University of South Africa, South Africa

Dr Patience Erick, Department of Environmental Health & Department of Public Health, University of Botswana, Botswana

Prof George Osanjo, Faculty of Health Sciences, University of Nairobi, Kenya

Lived experience advisory group (LEAG)

Purpose: Advisory board of experts with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. Drawing on their unique expertise and wisdom, LEAGs provide vital community-centred perspectives and guidance that inform the overarching approach and outputs of the project.

Members: Josephine Ewu, Yenagoa, Nigeria

Dr Tata Elvis, PRF Foundation, Cameroon

Timipre Sunday, Nigeria

Martin Bangura, Sierra Leone

Lisa Mudoya, Zimbabwe

Youth ambassador (YAs)

Purpose: Youth advisors (aged 18-29) with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. YAs bring unique youth-centred perspectives to the development and implementation of project activities.

Members: Lindokuhle Sibiya, Eswantini

Methodology

We produced this research and action agenda through a robust and inclusive methodology to capture, combine and refine a rich diversity of perspectives while fostering connection across a growing community of practice.

The CCM core team developed this methodology in consultation with RCTs, a Global Advisory Board and Wellcome. Methods and materials were adapted regionally to ensure a balance of global standardisation with regional appropriateness and flexibility. Continuous sharing between regions of processes, learnings and challenges facilitated the iterative development of the methodology. The process for developing the regional research and action agendas is shown below (Figure 1).

Figure 1: Research and action agenda development methodology



Pre-dialogue scoping

Global scoping and framing

- We performed a global-level scoping review of current reviews, key papers and policy reports relating to the climate change and mental health field.
- We undertook a mapping of the research categories covered by previous reviews and literature.
- We conducted a second mapping of the recommendations for action on climate change and mental health proposed by other reports.
- We used these results to frame the dialogues and research agenda to align with the current field while responding to key gaps.

Regional scoping

- We engaged with lived experience experts through key informant interviews to ensure that the production of the research agenda was appropriately grounded in an understanding of the key lived experience needs emerging in the region.
- We undertook oral reviews through interviews with climate scientists and mental health experts in Uganda, Cameroon, Nigeria, Ghana, Eswantini, South Africa, Botswana, Kenya, Gabon, DR Congo, Togo, Benin, Niger, Mauritania, Carbo Verde, Senegal, Ethiopia and Central African Republic. We also undertook targeted searches in PubMed, Google Scholar and Web of Science for literature reviews, scoping reviews and publications on research themes, priorities and gaps using “climate change impact on mental health” as search themes and search key words. A dearth of peer-reviewed papers specifically addressing climate change impacts on mental health was noted; the topic is mostly mentioned in newspapers, print and radio spaces.

Dialogue methodology

- We held two virtual dialogues (3 hours each) with experts across disciplines, sectors and countries in SSA.
- Dialogue discussions were designed by the global team, amended to be locally appropriate, and facilitated by members of the RCoP, who are themselves research, policy and lived experience experts.
- Many facilitators actively contributed to the discussions and therefore played a role in both data collection and production.
- The first dialogue identified regional needs and generated research priorities.
- The second dialogue gathered feedback on draft research priorities, identified how to enact research in the region and translate evidence to action in policy and practice, and explored the diversity of regional perspectives and understandings of key relevant concepts.
- The spectrum mapping exercise covered concepts related to mental health in the African context, climate change in the African context and African Indigenous knowledge on climate change and mental health. Concepts included stigma, underexplored mental health challenges, lack of awareness and Ubuntu approaches.
- Data generated in the dialogues included: Google Jamboard notes written by participants, Zoom chats, notes made by dedicated notetakers and transcripts of all discussions.

Dialogue agendas can be found in Appendix 1.

Survey methodology

Surveys (online)

- We distributed a **pre-dialogue survey** prior to Dialogue 1 to inform dialogue design and to solicit perceptions on regional climate impacts, climate-related mental health impacts and research priorities.
- We distributed a **post-dialogue survey** after Dialogue 2 to obtain a second round of feedback on research priorities and to identify relevant methods, metrics and datasets to address these priorities. This led to the inclusion of some additional questions in the final survey.

Full details of the survey methodology are outlined in Appendix 2.

Analysis methodology

- **Research categories:** The Climate Cares Centre conducted a global landscaping exercise of relevant existing climate change and mental health reviews and identified four broad research categories as areas of critical need for further work globally. This framework was used as the basis for structuring discussions within dialogues to generate research priorities and formed the global coding framework for analysis.
- **Analysis:** Dialogue data (transcripts of breakout rooms and notes) was analysed using the Framework Method – a matrix-based approach that allows qualitative researchers to undertake deep interrogation of transcripts and written notes. The analyst team populated a matrix based on the global coding framework by creating nuanced summaries with key quotes drawing from the notes and transcripts of the break-out discussions.
- **Draft priority research themes:** This matrix was used to draft a list of priority research themes, refined through consultation with the RCoP and triangulation across breakout notes, transcripts, pre-dialogue scoping and expert consultation. Selection of priority research themes adhered to a globally developed structure and selection criteria (e.g., potential to answer greatest regional emerging needs and evidence gaps, potential to inform decision making in policy and practice, research feasibility). See Appendix 3 for more details.
- **Refinement of priority research themes:** Research themes were shared with Dialogue 2 participants who gave feedback in response to the following prompts under the respective research themes:
 - Creating knowledge through research*
 - How can we best implement the research agenda and take forward research at the intersection of climate change and mental health in this region?
 - Fostering evidence-based policy and action*
 - How can we translate the growing evidence base to action in policy and practice?
- Research themes were refined in response to this feedback and shared with dialogue participants and a wider sample of experts in the post-dialogue survey. Participants gave free text responses to the question '*Is there anything you would like to change, remove or add to this research theme?*' and were given the option to suggest additional research themes.
- **Finalisation of priority research themes:** A final list of priority research themes was generated based on incorporation of survey feedback, consultation with the RCoP and regional experts, CCM core team, Global Advisory Board and Wellcome.

Coding frameworks for the research agenda and action agenda can be found in Appendix 4.

Participants

Dialogue participants were a diverse group across geographical spread, gender, sector, and discipline. All participants were invited to both dialogues, however in some cases participants were unable to attend both dialogues.

In total 46 participants attended Dialogue 1 and 51 participants attended Dialogue 2. The tables below provide a breakdown of participant characteristics.

Geographical spread

Country	Dialogue 1		Dialogue 2	
	Number	Percentage	Number	Percentage
Botswana	1	2.60%		
Cameroon	1	2.60%	1	3.4%
Eswatini	1	2.60%	1	3.4%
Guinea	1	2.60%		
Haiti	0	0%	1	3.4%
Kenya	10	26%	10	34.5%
Nigeria	6	16%	7	24%
Sierra Leone	3	8%	1	3.4%
South Africa	6	16%	3	10.3%
Uganda	3	8%	1	3.4%
United Kingdom of Great Britain and Northern Ireland	2	5%	3	10.3%
United Republic of Tanzania	1	2.60%		
Zimbabwe	3	8%	1	3.4%

Expertise

Expertise	Dialogue 1		Dialogue 2	
	Number	Percentage	Number	Percentage
Climate Change	23	30%	16	30.8%
Mental Health	18	23.4%	12	23%
Health	24	31.2%	15	29%
Other	11	14%	8	15%

I do not know / Prefer not to say	0	0%	0	0%
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Discipline

Discipline	Dialogue 1		Dialogue 2	
	Number	Percentage	Number	Percentage
Activism	18	13%	13	13%
Community	13	10%	10	10%
Education	19	14%	14	14%
Expert through my own lived experience	13	9.6%	10	10%
Funding	3	2%	3	3%
Healthcare	12	9%	9	9%
Non-governmental Organisation	13	9.6%	10	10%
Policy	12	9%	7	7%
Research	30	22%	20	20%
Other	3	2%	3	3%

Gender

Expertise	Dialogue 1		Dialogue 2	
	Number	Percentage	Number	Percentage
Men	14	37%	11	38.0%
Women	21	55%	14	48.2%
Non-Binary	1	2.6%	1	3.4%
I do not know / Prefer not to say	1	2.6%	2	7.0%
Other	1	2.6%	1	3.4%

Number of survey participants:ⁱ

Pre-dialogue survey: 90

Post-dialogue survey: 18

Ethics, data collection and storage

This study has been reviewed and given an ethical favourable opinion by the Imperial College Research Ethics Committee (study title: “Global Dialogues to set an actionable research agenda and build a community of practice in climate change and mental health”; study ID number: 6522690) and locally by the Foundation for Professional Development Research Ethics Committee (reference number: REC-03711-033-RA).

The study was conducted in accordance with the recommendations for physicians involved in research on human subjects adopted by the 18th World Medical Assembly, Helsinki 1964 and later revisions, as well as in alignment with the South African Protection of Personal Information Act (POPIA) of 2020.

Details on data collection and storage can be found in Appendix 5.

Current state and emerging needs for climate change and mental health in Sub-Saharan Africa

As the climate crisis escalates, more people globally are experiencing related mental health consequences. However, the current evidence base doesn't fully capture these experiences. To develop research themes that ultimately meet the needs of those experiencing and responding to the interconnections between climate change and mental health, it is vital to know: 1) what do people from different backgrounds, contexts and sectors – particularly those with lived experiences of mental health challenges in the context of the climate crisis – report as their experiences, needs and resiliencies, and 2) what evidence do people making decisions and taking actions on the ground need in order to adequately respond?

This section sets out the context of the research agenda, presenting a synthesis of what we heard through the dialogues, surveys, expert consultations and literature review as key emerging needs for mental health in the context of climate change. We also present regional perceptions and insights from the literature on the current state of evidence to explore which key emerging needs are not yet being captured through research.

ⁱ Please note numbers are approximate and do not account for duplicate or incomplete responses.

Current research on climate change and mental health in Sub-Saharan Africa

Climate change can have a significant impact on mental health, particularly in vulnerable regions such as SSA. The region is already facing numerous challenges, including poverty, political instability and inadequate healthcare; climate change exacerbates these issues. Based on existing literature, climate change can affect mental health due to:

- *Extreme weather events*: Increased occurrence and severity of extreme weather events – such as storms, droughts and floods – can cause displaced people to lose their means of subsistence as well as create worry and anxiety. This accompanying anxiety and uncertainty may have long-term psychological impacts.¹²
- *Food insecurity*: Climate change impacts agriculture, with changes in rainfall patterns and temperature affecting crop yields. This can result in food shortages, malnutrition and heightened stress levels, contributing to poor mental health.¹³
- *Water scarcity*: Variations in precipitation can cause water shortages, which impact the water available for agriculture and drinking. Lack of access to clean water can exacerbate illnesses linked to poor sanitation, which can affect physical health and may in turn increase the prevalence of mental illness.¹⁴
- *Vector-borne diseases*: Climate change can influence the distribution of diseases such as malaria and other vector-borne illnesses. The increased prevalence of these diseases can lead to more health concerns and stress for individuals and communities.¹⁵
- *Conflict and migration*: Geopolitical disputes over land, water and other vital resources are a result of climate change and resource shortages. Communities can be upended by migration brought on by climate change, and displaced people may experience mental health issues as a result.¹⁶
- *Economic stress*: Climate change can affect economic sectors such as agriculture and fisheries, leading to job loss and economic stress. The resulting economic insecurity can contribute to mental health issues, including depression and anxiety.¹⁷
- *Inadequate healthcare*: Many countries in SSA already face challenges in providing adequate healthcare. Climate change can strain healthcare systems further, making it difficult for people to access support.¹⁸

Regional needs for mental health in a changing climate

While the climate crisis affects every part of the world, dialogue outcomes suggest that current and future hazards are unevenly distributed and variable. For instance, coastal communities may face increasing rises in sea level, coastal erosion and storm surges. Different parts of the same country or region may face different levels of threat to droughts, wildfires, floods or extreme heat. Understanding the predicted climate hazards and their timescales for different communities and regions is vital to appropriately target support for mental health.

This section outlines the key climate exposures facing SSA projected to approximately 2030 as compared to historical baseline (generally 1986-2005), as modelled by climate experts at the Red Cross Red Crescent Climate Centre. We also present findings from the dialogues, surveys, and consultations with lived experience experts to explore perceptions of: 1) mental health risks associated with these climate exposures, 2) whose mental health may be most at risk, 3) the pathways through which climate exposures might produce or exacerbate existing mental health problems, 4) climate adaptation and mitigation actions that may benefit mental health, and 5) mental health actions/solutions that can help respond to these mental health impacts.

Climate hazards

SSA is one of the regions most vulnerable to climate change. The region has regularly experienced strong droughts, wildfires, floods, landslides, storms and extreme temperatures in recent years, according to the international disaster database EM-DAT. The region is facing an increase in the frequency and severity of a range of climate hazards, including the following, modelled to approximately 2030 as compared to historical baseline (generally 1986-2005):^{19, 20}

- **Mean and extreme temperatures**, including more than 40 additional hot days per year above 35°C for the Horn of Africa, western African Coast and southern Africa (high confidence);ⁱⁱ
- **Droughts and wildfires** in the Southern Africa region, though there remains uncertainty across the range of global climate models (low to medium confidence);
- **Heavy precipitation events and flooding**, with the frequency and intensity increasing almost everywhere in Africa with additional global warming (high confidence);
- **Sea-level rise**, contributing to increases in the frequency and severity of coastal flooding in low-lying areas and to coastal erosion along most sandy coasts (high confidence); and
- **Tropical cyclones**, particularly the proportion of Category 4-5 tropical cyclones in Eastern Africa (low confidence), though the spread (indicating the range

What mental health outcomes appear to be impacted?

Anxiety, depression, PTSD and suicide were identified by dialogue participants as climate-related mental health issues in SSA. These issues can be indirectly caused by climate change and can arise from impoverishment, long-term drying in rural areas, family member loss, injury that renders one unable to work or relocation.¹ However, more research is needed to understand the specific mental health challenges arising from climate change in SSA. This includes studying the social, economic and cultural factors which influence vulnerability and resilience.

Who is particularly affected?

Dialogue participants identified a number of vulnerable communities as being particularly affected, including those listed below.

- Displaced people, such as migrants, asylum seekers and climate refugees
- Fisher peoples
- People with a pre-existing mental health or health condition
- Persons with disabilities
- Subsistence farmers situated in drought and fire-prone regions
- The elderly
- The poor living in informal settlements
- Those living in rural areas
- Those living in high-risk areas such as at the coast
- Women (particularly those living in drylands and arid/semi-arid regions)
- Youth

ⁱⁱ Confidence levels listed after projections are a measure of researcher confidence that model's results are true and not due to chance.

Building resilience at the community level is crucial. This involves not only addressing immediate mental health needs, but also empowering vulnerable communities to cope with and adapt to long-term impacts of climate change.

What are the pathways and mechanisms linking these climate hazards to mental health outcomes?

Dialogue participants noted that many sub-Saharan African nations struggle with issues related to mental healthcare, such as stigma, insufficient funding and poor infrastructure. Climate change compounds these challenges by introducing additional stresses and threats to livelihoods. The literature indicates that increases in the frequency and intensity of extreme weather events, such as droughts, floods and storms, can lead to displacement, loss of property and economic instability. Changes in climate patterns can also negatively affect agriculture, leading to food shortages and malnutrition.²¹ Dialogue findings suggest that food insecurity exacerbates mental health conditions, such as anxiety and depression, in addition to detrimental effects on physical health. Climate-related water scarcity can result in the unequal distribution of water resources as well as shifts in patterns of availability. In addition to having an adverse effect on physical health, a lack of access to clean water can also cause mental health challenges.

What climate adaptation and mitigation actions have benefits for mental health?

Dialogue results suggest that Indigenous and local knowledge provides many African communities with coping mechanisms to address the negative effects of climate change and to adapt to environmental stresses. These include weather-forecasting systems based on changes in animal behaviour, livestock and crop diversification, cattle stress-management techniques and division of labour, thus allowing acclimatisation to drought challenges; this acclimatisation can reduce individual stress and benefit mental health, outcomes of which are also supported in the literature.²²

What mental health actions can help respond to these mental health impacts?

Dialogue results indicate that strengthening the capacity of healthcare systems to address mental health issues related to climate change is the most critical adaptation and mitigation consideration within the sub-Saharan context. This includes training healthcare professionals, creating awareness, extensively mobilising community-based lay counsellors and integrating mental health services into primary healthcare, outcomes of which are also supported in the literature.¹⁸ Addressing mental health impacts in SSA further involves a multifaceted approach that includes a combination of self-care, Indigenous healing practices (which include Ubuntu approaches and nature-based interventions), professional support and community resources.

Further contextual information on regional needs

With increasing climate devastation on the sub-continent, many affected communities in SSA are yet to fully measure their mental health burden due to already existing mental health challenges linked to cultural taboos and socioeconomic challenges. It is clear from the dialogues that the methodologies for diagnosis of mental health challenges and metrics need to be carefully studied in SSA. Baseline mental health challenges and morbidity may be slightly different to other continents given, for example, that SSA is more prone to epidemics and pandemics. Due to the global nature of climate change, international

support and collaboration is also essential. This may include financial assistance, technology transfer through capacity building and knowledge sharing to help sub-Saharan African countries build resilience and adapt to changing climate conditions.

Research agenda

Priority research themes

Background to research categories and priority research themes

This research agenda presents an aligned vision to guide the climate and mental health field in SSA. Research priorities have been generated through consultation with experts across disciplines, sectors and geographies in the region and iterated with experts regionally and globally. The priority research themes represent areas where targeted research investment could create a full picture of impacts, their mechanisms, and solutions across both mental health and climate actions. We outline why these have been identified as priorities and how they can be addressed by combining expertise across disciplines and sectors.

Research priorities are presented within four overarching research categories that were identified as areas of critical need for further work globally and that map the climate and mental health research space at a high level, based on an initial review of global literature. Note that some priorities span multiple categories.

- **Impacts, risks and vulnerable groups:** improving our understanding of the ways in which mental health is affected by climate change. For example: what mental health outcomes are impacted or at risk; the prevalence, severity, economic and societal costs of these impacts; and who is most vulnerable to these impacts.
- **Pathways and mechanisms:** improving our understanding of how mental health is affected by climate change and, in particular, whether there are factors specific to climate change that increase mental health risks. This includes considering biological, psychological, societal or environmental pathways and mechanisms.
- **Mental health benefits of climate action (adaptation and mitigation):** understanding and quantifying when and how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health.
- **Mental health interventions/solutions in the context of climate change:** identifying the most effective mental health interventions/solutions/actions to support mental health in the context of climate change, across diverse sectors. This encompasses providing support to people already experiencing negative mental health impacts and reducing risk or severity of future negative mental health impacts.

We hope that these priorities will act as an inspiration to guide to the direction of the research community of practice, investment targets for funders, and the evidence base for policy makers and practitioners to address the emerging and predicted mental health needs in response to climate change in SSA.

Priority research themes

Research category	Priority research theme (n = number of research questions)
1. Impacts, risks and vulnerable groups	1. Understanding and quantifying the nature and magnitude of the mental health burden in relation to climate change in SSA. 2. Understanding how mental health stigma, historical trauma (for example, the long-term mental health impacts of apartheid in South Africa) and inequalities affect access to care within the context of the nexus between mental health and climate change in SSA. In terms of mental health stigma within the SSA context for example, it is important to understand how stigma may influence 1) climate-related mental health outcomes, 2) design of climate-related mental health interventions, and 3) access to climate-related mental health interventions. 3. Identifying the vulnerable groups most at risk of climate change-related mental health challenges. <i>Proposed vulnerable groups for further study include: the poor living in informal settlements, Indigenous populations, those living in rural areas, people with a pre-existing mental or physical health condition, people with disabilities, women (particularly those living in in drylands, arid/semi-arid regions and coastal areas prone to flooding and rising sea levels), youth, the elderly, those living in high-risk areas such as at the coast, (subsistence) farmers situated in drought and fire-prone regions, fisher peoples and those displaced such as migrants, asylum seekers and climate refugees.</i> 4. Understanding whether vulnerable groups in SSA hold any cultural perceptions or dogmas/taboo with regard to climate-related mental health challenges and, if so, how these impact mental health outcomes and interventions. 5. Determining the most relevant standardised tools, indicators and methodologies to measure the effects of climate change on mental health within the sociocultural and socioeconomic context of SSA.
2. Pathways and mechanism	1. Identifying the direct causal pathways between climate change and mental health (for example, if direct experience of extreme heat worsens mental health outcomes) within the sub-Saharan context. 2. Identifying the indirect causal pathways between climate change and mental health within the sub-Saharan context (for example, if there are any HIV-associated neurocognitive disorders [HAND] symptoms - such as problems with mood, concentration, irritability - that can be exacerbated by extreme heat, or whether and how heat-exacerbated aggression is an indirect pathway affecting mental health outcomes). 3. Exploring the syndemic, co-morbid and spill-over relationships between climate change-related mental health challenges and other stressors on mental health such as wars, violence, trauma and epidemics (for example, the geopolitical conflicts in Nigeria, HAND, Covid-19 experienced across the region and the Ebola crises in West Africa). 4. Identifying and understanding the climate change-related mechanisms that increase the risk of mental health challenges within population subgroups in SSA. For example, identifying whether there are any specific risks, triggers and stressors related to

climate change that increase the risk of mental health challenges for certain vulnerable groups.

3. Mental health benefits of climate action (adaptation and mitigation)

1. Identifying the existing coping mechanisms and resilience strategies employed by sub-Saharan communities in response to climate-related stressors and their effects on mental health.
2. Assessing the impact of establishing weather warning technology on disaster mitigation in SSA and whether this might improve outcomes for climate-related mental health challenges.
3. Evaluating whether resettlement plans for climate-related displaced communities enhance their mental health in countries such as Sierre Leone.
4. Exploring how lived-experience communities can meaningfully co-create effective climate change and mental health policies and support their implementation in sub-Saharan countries.
5. Understanding whether accelerating climate action via climate justice can improve climate-related mental health outcomes in SSA.
6. Assessing the level of political buy-in of governments regarding climate change and its associations with mental health in SSA and how this affects the co-beneficial actions taken by governments.
7. Understanding whether the advancement of sustainable supply chains (in all domains, both direct and indirect) mitigates the impacts of climate change on mental health in SSA (for example, advancing sustainable practices in food systems in ways that reduce job losses in the agricultural sector and prevent food insecurity, thus diminishing the likelihood of presentations of mental health challenges).
8. Understanding which climate adaptation or intervention strategies – such as methods of post-disaster recovery, including pandemic preparedness and response initiatives – can be adapted to also address climate-related mental health challenges within SSA.

4. Mental health interventions/solutions in the context of climate change

1. Evaluating the efficacy of African Indigenous medicines, counselling techniques or practices in treating climate change-related mental health issues.
2. Exploring whether drawing on a culture of 'ubuntu' (a collectivistic humanist African world view, meaning 'I am, because you are') can aid psychosocial support and improve climate-related mental health outcomes.
3. Evaluating the efficacy of implementing a mental health task-shifting approach within the SSA context, given the heightened burden that climate change will impose on the mental health system.
4. Assessing whether technologies (such as telemedicine/online counselling, gaming, AI products, low technological applications such as messaging apps, offline technologies and low data solutions) can be used to provide effective climate-related mental health support and increase access to care within SSA.

5. Identifying and evaluating strategies to ensure that international agencies, country-specific government agencies, academia, employer groups, unions and NGOs effectively collaborate to support climate change and mental health interventions in SSA.
6. Understanding ways to integrate climate change adaptation strategies within mental health and health services (for example, establishing climate resilient health systems) to enhance resilience in SSA.
7. Measuring the prevalence of psychosocial help-seeking for climate distress and climate-specific mental health or wellbeing challenges, (e.g., PTSD and eco-anxiety), in SSA and its relationship with mental health outcomes. Understanding whether those seeking care for mental health challenges (e.g., PTSD) or climate-related distress and wellbeing challenges (e.g., eco-anxiety) have access to care and whether access can be expanded, if necessary.
8. Assessing the effectiveness of awareness campaigns (for example, social media awareness) in supporting mental health in the context of climate change and encouraging the use of appropriate services in SSA.
9. Assessing the effectiveness of peer support initiatives (for example, a crisis hotline) in supporting mental health in the context of climate change and encouraging the use of appropriate services in SSA.
10. Evaluating strategies to enhance the climate-related mental health resilience of individuals and communities in SSA.
11. Evaluating the need to adjust mental health clinical diagnostics (for example, ICD-10 and DSM-5-TR diagnostics systems) and practices to incorporate climate change-related mental health pathologies, such as severe and debilitating climate anxiety, within SSA.
12. Assessing the readiness of the mental health workforce and healthcare system to deal with climate change-related mental health challenges in SSA.
13. Assessing how mental health system strengthening can be integrated into large climate change adaptation projects in SSA.
14. Evaluating and documenting best practices for integrating climate change and mental health curricula into training for healthcare professionals in higher institutions of learning in SSA.

Overview of themes for impacts, risks and vulnerable groups

Why were these themes chosen as priorities?

These themes are grounded in the recognition of the region's susceptibility to the adverse effects of climate change and the potential consequences for mental health. The selection of these themes was influenced by a number of factors, including the focus in the dialogues on climate-related effects on livelihoods and the consequences for mental health, the vulnerability of SSA to climate change, the stigma associated with mental health (including “double stigma” associated with living with HIV and having mental health issues), the presence of vulnerable groups in the area, the limited capacity for

adaptation and the need to understand how contextualised and culturally sensitive therapeutic interventions can mitigate and address the impacts on mental health. The need to develop standardised tools, indicators and methodologies was also highlighted.

“Mental health is not really understood within the African context and that which is understood is more the extreme cases – for example, psychotic cases or schizophrenia.” (Participant, Dialogue 1, Breakout session 1)

“It's the same thing as when we discovered HIV-AIDS. Once people knew that you were HIV-positive, then no one will come close to you. It's the same as mental health. People see you as a different animal. You need to be isolated from society.” (Participant, Dialogue 1, Breakout session 1)

What are examples of methodologies, metrics and datasets that could be used to address these?

Several mental health and climate change datasets and metrics were identified in the dialogues. A key next step – across all research categories – would be to initiate a more comprehensive scoping review of relevant datasets in the region to identify relevant data or opportunities to collect such data in the future with key partners. Further investment in creating guidance for key stakeholders (such as governments and research institutions) in the region to routinely collect relevant data is also warranted.

Table 3: Methodologies, metrics and datasets for impacts, risks and vulnerable groups themes

Methodologies	Metrics	Datasets
Climate modelling	Adaptive capacity indexes	Case studies
Clinical observations	Climate vulnerability indices	Climate data
Community-based participatory research	Health service accessibility indexes	Community and household surveys
Cross-sectional surveys	Mental health and mental wellbeing metrics (e.g., the Warwick-Edinburgh Mental Wellbeing Scale)	Demographic and socioeconomic data
Data mining	Resilience indicators	Epidemiological studies
Focus group discussions		Geospatial data
Health impact assessments		Health surveillance data
Qualitative research		Longitudinal studies
Structured questionnaires		Satellite imagery
Vulnerability assessments		

Overview of themes for pathways and mechanisms

Why were these themes chosen as priorities?

Understanding the direct and indirect pathways and mechanisms of climate change and mental health in SSA will provide a nuanced understanding of the challenges faced by the region, with the ultimate goal of promoting resilience and sustainable development in vulnerable communities.

“We also have to look at the direct and indirect research, so it's your indirect impacts and also your direct impact. For example, heat is known, but there isn't a lot of research, but there's some research that heat can heighten aggression. Then there's also a lot of indirect pathways. If I am stressed because of a flood, I could have experienced a great deal of

trauma, so then looking at PTSD and those kinds of things". (Participant, Dialogue 1, Breakout session 2)

Participants highlighted numerous reasons for choosing these priority themes, including the interconnected nature of climate change and mental health, and the unique vulnerabilities experienced by SSA. These include the potential for amplified risks due to existing social and economic inequalities, and the need to advance health equity and social justice. Understanding the syndemic, co-morbid and spill-over pathways through which climate change affects mental health is essential for anticipating and preventing potential mental health crises. Identifying direct and indirect pathways and mechanisms may provide a foundation for developing targeted, preventative and adaptive strategies for the region.

What are examples of methodologies, metrics and datasets that could be used to address these?

Table 4: Methodologies, metrics and datasets for pathways and mechanisms themes

Methodologies	Metrics	Datasets
Case studies	Mental health indices	Climate data
Engaging communities through participatory research methods and platforms (e.g., WhatsApp)	Social capital indicators and adaptation metrics	Climate impact assessments
Focus group discussions	Stress and resilience metrics	Community and household surveys
Interviews		Geospatial data
Longitudinal studies to track changes in mental health indicators over time		Health surveillance data
Population-based studies to assess the prevalence and distribution of mental health disorders in relation to climate change-related stressors		Remote sensing data
		Satellite imagery
		Socioeconomic datasets

Overview of themes for mental health benefits of climate action [adaptation and mitigation]

Why were these themes chosen as priorities?

Dialogue participants noted that adaptation and mitigation strategies to address climate change can be beneficial to mental health in SSA. Further, the mental health advantages of climate action are linked to more general development objectives, creating greater momentum for change and policy integration. By integrating mental health considerations into climate policies and adaptation strategies, there is an opportunity to create more holistic and resilient communities in SSA and beyond.

"...it's looking at what do countries have in their constitutions? What do they have in their basic legislation? And what is in there that you can point out and say, this is the case I can make that in the current policy, and the guidelines...are things that are supporting mental health, it supports the climate change agenda around mental

health, as well.” (Participant, Dialogue 2, Breakout session 1)

What are examples of methodologies, metrics and datasets that could be used to address these?

No specific datasets were identified in the dialogues. As above, further research into appropriate methodologies, metrics and datasets relating to climate-mental health in SSA is a necessary first step.

Table 5: Methodologies and metrics for mental health benefits of climate action (adaptation and mitigation) themes.

Methodologies	Metrics
Climate vulnerability assessments	Adaptive capacity indicators
Community-based participatory groups	Climate vulnerability indices
Epidemiological studies	Environmental impact metrics
Health impact assessments	Mental health metrics
Social vulnerability assessments	Resilience indicators
Talking circles among youth groups	

Overview of themes for mental health interventions/solutions in the context of climate change

Why were these themes chosen as priorities?

These themes reflect the substantial impact of environmental changes on mental wellbeing and the necessity for comprehensive, Indigenous, long-term strategies to support affected populations in SSA. Some participants highlighted the advantage that African countries have over Westernised countries, for example, by adopting more collectivist and communal lifestyles and how these principles could be applied to mental health interventions in the context of climate change.

“The other issue is that of Indigenous knowledge and preservation and conservation. For example, if you talk about medicinal plants, if you look at climate and the drought, you may have elimination of some plants that were used in African herbal practice in treatments of mental illness.” (Participant, Dialogue 1, Breakout session 1)

“We have advantages compared to the European context. One, we are largely communal. It's only the Africans that are now maybe living in peri-urban and urban settings that are losing the sense of community, but we are largely communal. So, with the right information and with the knowledge and information on mental health, we have an advantage of being there for the people that need support, especially the psychosocial support, over the European context.” (Participant, Dialogue 1, Breakout session 1)

“A lot of the problems in the so-called West [are] because of the isolation that people feel because it's very individualistic and not communal or ubuntu-based like Africa is. It's a whole new area of enquiry that needs to evolve out of Africa of how do we look at these things in a unique way, that has substantial advantages over the traditional Western perspective. I'm not negating the role of psychiatric medications. They are needed in certain places. I do think they overprescribe. That's a personal view...One can look at what other methodologies. Just for interest's sake, I was speaking to a chief psychiatrist in the UK. They were talking about nature-based therapy, what role can nature play in healing trauma, for example.” (Participant, Dialogue 1, Breakout session 1)

Other priorities stemming from the dialogues include how culturally-sensitive and inclusive interventions are critical in humanitarian and developmental contexts to support individuals and communities in coping

with trauma, stress and loss. Furthermore, leapfrogging to telemedicine/telecounseling and other technologies that offer offline and low data solutions, for example, may bridge some of the shortfalls in the shortage of human resources in health in the region.

"We mustn't underestimate the role of lay counselling and telemedicine." (Participant, Dialogue 1, Breakout session 2).

It is also important to determine the readiness of the relevant workforce and healthcare system in sub-Saharan Africa and the availability of mental health services to deal with any climate change-related impact.

"Human resources for health [are] obviously an issue there, but there could also simply be issues regarding availability, for example. That includes availability of commodities for health, which includes drugs. For example, are medicines available? Even the mental health services by themselves, maybe the infrastructure. Do we have centres, apart from the human resources for health. Maybe, yes, perhaps a very broad matter that requires further subdividing into smaller and smaller issues that can be researched. It's big." (Participant, Dialogue 1, Breakout session 2)

"Africa doesn't have sufficient mental health practitioners. There aren't enough psychiatrists, psychologists, social workers, so what one could consider is looking more at a continuum. There's actually a public health term for it... task shifting.... So looking at the task, doing research on task shifting approaches. So that means, for example, instead of a psychiatrist or a psychologist helping somebody, because we don't have enough in Africa and it's simply not realistic, you'd have to look at a task shifting approach. For example, teaching nurses or auxiliary nurses to be the frontline people. It can also be environmental health practitioners, auxiliary social workers." (Participant, Dialogue 1, Breakout session 2)

What are examples of methodologies, metrics and datasets that could be used to address these?

Table 6: Methodologies, metrics and datasets for mental health solutions/interventions in the context of climate change themes

Methodologies	Metrics	Datasets
Community-based participatory research	Climate adaptation and mitigation metrics	Climate action implementation data
Mixed methods research	Climate vulnerability indices	Community surveys
Participatory action research	Health-related quality of life metrics	Economic and livelihood data
	Psychological wellness scales (e.g., the Warwick-Edinburgh Mental Wellbeing Scale or the Connor-Davidson Resilience Scale)	Environmental monitoring data
	Social cohesion indicators	Health surveillance data
	Stress and anxiety indices	Longitudinal data
		Remote sensing and geographical information system data

Action agenda

As the climate change and mental health field in SSA builds and the evidence base grows, it is crucial to avoid perpetuating existing challenges, including: 1) disconnections across disciplines and between researchers and policymakers; 2) unequal focus on topics and geographies; and 3) and siloed decision making for climate and for mental health. This action agenda sets out a shared vision as a rallying focus of the mental health and climate change field in SSA. This includes guidance on how to best support the growing community of practice, how to translate evidence to action, and the principles that should guide this approach. Enacting this agenda will require transdisciplinary effort and coordinated action across research, research funding, policy, and practice. This action agenda aims to guide this work by setting out the challenges that must be addressed, opportunities that can be harnessed, and priority actions to work towards a thriving climate and mental health field.

Regional vision for mental health in a changing climate

Vision statement

We envision a future where mental health is prioritised, resilient communities thrive and individuals can cope effectively with the challenges posed by a changing climate. Our vision is grounded in inclusivity, cultural sensitivity and evidence-based practices, fostering a holistic approach to wellbeing that integrates mental health into climate change adaptation and mitigation strategies.

Creating an enabling environment for research at the intersection of climate change and mental health

The identified research priorities will only be of value if they are enacted. The climate and mental health field is relatively new and rapidly growing, and now is the time to ensure that it is designed to deliver a mentally healthier future in the context of the climate crisis. In a field that spans multiple disciplines and sectors, each with different cultures and ways of working, and on a topic with low awareness in many countries, what is needed to support capacity-building efforts? What principles must guide the field, and what are the challenges and opportunities in the region to create an environment that would enable such research?

This section presents a synthesis of dialogue discussions and survey results on what is needed to implement the research agenda and foster an enabling research environment for climate change and mental health.

A desired state of climate change and mental health research in Sub-Saharan Africa

SSA's ideal future state for climate change and mental health supports wellbeing, resilience and sustainable development. Key characteristics include:

- Providing integrated approaches for mental health services;
- Extensively training lay counsellors (schooled in in interpersonal counseling, for example);
- Fostering cultural sensitivity;
- Building community resilience;

- Promoting education and awareness campaigns;
- Integrating climate change and mental health policy;
- Advancing cross-sectoral collaboration;
- Creating early warning systems; and
- Organising international cooperation.

Achieving this desired future state will require a concerted effort from governments, communities and international partners. Rooting out corruption in many governments was also cited as a key aspect that needs to be addressed in many SSA countries.

Challenges holding back research

Research related to the intersection of climate change and mental health in SSA faces several challenges, including:

- limited funding, which may hinder the scale and depth of research studies;
- vested interests and state corruption;
- misinformation;
- data gaps;
- capacity and infrastructure deficits;
- a lack of integration of climate change and mental health research;
- cultural and linguistic diversity (which may lead to studies which are culturally insensitive);
- African voices not being sufficiently heard on the global stage;
- extractivist research being conducted by the Global North in SSA;
- causal pathways between climate change and mental health remain vague;
- political instability, which can disrupt research activity and hinder collaboration between researchers and policy makers;
- limited public awareness and education, which may hinder support for research initiatives and intervention; and
- lack of suitably adapted research methodologies.

Addressing these challenges requires a collaborative effort involving governments, non-governmental organisations (NGOs), researchers and local communities. This effort must include investing in education, building research capacity, promoting inter- and transdisciplinary collaboration, and integrating mental health considerations into climate change adaptation strategies.

Opportunities and enablers

We identified several opportunities to enhance research related to climate change and mental health, as summarised below.

- **Leveraging local knowledge** (for example, many African universities have adopted decolonial approaches in psychology curricula, while Deborah Adenikinju has developed a pan-African [Mental Health Map](#) which looks at resources and the presence of mental health work in various counties in the region) and **building on traditional coping mechanisms** to enhance community resilience to the mental health impacts of climate change.
- **Aligning efforts to address climate change and mental health within the sustainable development goals (SDGs)** to provide a framework for comprehensive, integrated intervention.
- **Integrating mental health considerations into climate action.** Promoting climate-smart agricultural practices, for example, not only contributes to climate change mitigation and adaptation, but can also positively impact the mental wellbeing of farmers; meanwhile, research investigating

sustainable agricultural practices can enhance food security and reduce stress associated with climate-related uncertainties.

- **Promoting the concept of One Health**, a multidisciplinary approach that acknowledges that the health of people is closely linked to the health of animals and the environment.
- **Promoting grassroots activism** and exploring (preventative) therapeutic approaches for existential trauma.

Relevant potential partners

Examples of suggested stakeholders for implementing a research agenda are outlined below.

- Africa Centres for Disease Control and Prevention (CDC)
- Community based organisations
- Disaster relief agencies
- Global Mental Health Action Network
- Governments
- International research funders (for example, Wellcome, National Institute of Mental Health (NIMH))
- NGOs
- Research institutions
- United Nations International Children's Emergency Fund (UNICEF)
- World Health Organization (WHO)

Partnerships between investors, activists, policy makers, researchers and community stakeholders (particularly rural community and transgenerational community representatives) can facilitate the exchange of knowledge, best practices and innovative solutions to address the complex challenges of climate change and mental health.

Priority next steps/recommendations to investors and actors

A coordinated effort from investors, policymakers, researchers and community stakeholders is needed to create an enabling environment for research at the intersection of climate change and mental health in SSA.

Investors need to:

- allocate funds specifically for transdisciplinary and interdisciplinary research projects that explore the complex interaction between climate change and mental health in SSA;
- encourage grant programmes that support long-term studies;
- provide grants for capacity building in research institutions and communities – this includes training researchers, healthcare professionals and community leaders to conduct and utilise research effectively;
- foster public-private partnerships to leverage resources and expertise, and encourage collaboration between philanthropic organisations, private companies and governmental bodies to support comprehensive research programmes; and
- invest in technological solutions for data collection, analysis and dissemination, and support the development and implementation of innovative technologies such as mobile health applications and remote sensing tools.

Policy makers need to:

- advocate for the integration of mental health considerations in national and regional climate change policies and vice versa;
- invest in developing the climate-mental health evidence base to develop appropriate and effective policies;
- provide incentives for research institutions and organisations to collaborate in inter/transdisciplinary research in the field;
- create procedures for converting study results into practical policies; and
- set up legal frameworks that support moral and culturally aware research.

Researchers need to:

- actively seek collaborations with diverse disciplines, including climate science, psychology and public health;
- prioritise participatory research methods that respect local Indigenous knowledge and incorporate the perspectives of those directly affected by climate change and mental health challenges;
- commit to open data sharing and investing in capacity building for researchers; and
- encourage transparency in the research process, including informed consent, and ensure that research benefits are communicated to participants.

Community stakeholders need to:

- advocate for, and participate in, community-based research initiatives;
- engage in knowledge mobilisation activities to disseminate research findings within communities; and
- advocate for the participation of community members in research studies.

Translating a growing evidence base into action that can respond to the mental health impacts of climate change

The current evidence base on the interconnections between climate change and mental health compels action in policy and practice to protect escalating mental health needs and promote co-beneficial climate actions. How can current evidence and new insights created through implementing this agenda best translate into changes to policy decisions and practices across both climate and mental health regional spaces?

This section presents a synthesis of dialogue discussions and survey results, setting out the challenges and opportunities to translate evidence generated through research into policy and practice.

A desired state of climate change and mental health to action in Sub-Saharan Africa

The relationship between climate change and mental health in SSA is a complex and multifaceted issue. Achieving a desired state of climate change and mental health in the region involves addressing various pre-existing and endemic challenges, such as poverty derived from environmental degradation, and implementing evidence-based actions. Some key considerations include adopting ubuntu/collectivistic approaches in psychosocial support, renewable energy transition, water purification, organic agriculture and rural development, education and awareness.

Challenges holding back translation of evidence

Despite a growing evidence base on the mental health impacts of climate change in SSA, several challenges impede the translation of this evidence into effective action, including:

- limited mental health infrastructure;
- stigma surrounding mental health issues due to cultural belief systems and negative perceptions of mental health challenges;
- a potential false reliance on the concept of resilience;
- lack of awareness and understanding of the connection between climate change and mental health in both the general population and among healthcare providers;
- fragmented research efforts, with limited coordination and collaboration among researchers, policy makers and community stakeholders;
- gaps in data collection, particularly in remote or under-serviced areas;
- limited financial resources allocated to mental health initiatives and climate change adaptation efforts;
- lack of inclusion of mental health considerations in broader climate change policies and strategies;
- lack of community engagement in the development and implementation of mental health interventions; and
- the rapid pace of environmental changes outpacing the development and implementation of mental health interventions.

Opportunities and enablers

There are many opportunities and enablers to create a sustainable and resilient future which prioritises both environmental and mental health and wellbeing in SSA, as detailed below.

- **Fostering collaboration** between mental health professionals, climate scientists, policymakers and community leaders to develop comprehensive and effective strategies.
- **Establishing and expanding psychosocial support services** to help individuals and communities cope with the psychological impacts of climate change.
- **Strengthening mental health infrastructure** by training healthcare professionals, community outreach workers and community leaders to recognise and address climate-induced mental health issues.
- **Encouraging the use of nature-based solutions** and ecotherapy to contribute to mental health and wellbeing.
- **Connecting people with nature** and promoting approaches such as the [Friendship Bench](#).
- **Establishing early warning systems** for extreme weather events and climate-related stressors, allowing for timely interventions to protect mental health.
- **Utilising technology** – such as telemedicine and mobile applications – to provide mental health support to remote or underserved populations.
- **Promoting education about mental health (mental health literacy) to reduce the stigma** surrounding mental health issues and encourage individuals to seek help when needed.

Relevant potential partners

A vast range of stakeholders are needed to successfully translate evidence to action in SSA. Key partners and stakeholder groups are listed below.

- Community stakeholders
- Education sector
- Healthcare providers

- Investors and donors
- Policy advocates
- Policy makers
- Technology industry
- Youth representatives

Priority next steps/recommendations to investors and actors

In order to address the effects of climate change on mental health in SSA, we need coordinated action from policy makers, investors, healthcare providers, communities and other stakeholders.

Investors should:

- direct funding toward the implementation of evidence-based mental health intervention programmes;
- scale up successful interventions;
- invest in technology-enabled mental health solutions such as telemedicine, mobile apps and online /offline platforms;
- allocate resources for capacity building in mental health infrastructure; and
- support long-term studies that contribute to evidence-based practices tailored to the SSA context.

Policy makers need to:

- integrate mental health considerations into climate change policies and adaptation strategies and vice versa;
- allocate resources within national budgets for mental health services and programmes; and
- develop and implement guidelines for mental health training for healthcare providers and community workers.

Healthcare providers should:

- integrate mental health services into primary healthcare systems;
- conduct community outreach programmes to raise awareness about mental health issues related to climate change; and
- collaborate with NGOs to expand mental health services.

Community stakeholders should:

- empower local communities to lead mental health initiatives;
- use culturally appropriate communication methods to increase awareness and reduce stigma surrounding mental health;
- establish and strengthen community support networks; and
- advocate for the recognition of local knowledge and solutions.

By implementing these next steps, embedded in multi-sectoral and global partnerships, relevant actors can respond to the mental health impacts of climate change in SSA, fostering resilience and contributing to the overall wellbeing of the population.

Discussion: strengths, limitations and next steps for the research and action agenda

By harnessing strengths, addressing limitations and taking effective next steps, this research and action agenda can become a dynamic and responsive framework that effectively promotes mental health resilience in the face of a changing climate.

Strengths

- **Cultural relevance:** The agenda for research and action is tailored to the particular circumstances of SSA, taking into account the region's opportunities and challenges within the context of climate change and mental health.
- **Trans/interdisciplinary approach:** The agenda adopts a trans/interdisciplinary approach, involving climate scientists, mental health professionals, policymakers and community leaders. This holistic perspective enhances the effectiveness of proposed research priorities and action within the nexus of climate change and mental health.
- **Community engagement:** A focus on community involvement within the field of climate change and mental health ensures that the research and action agenda is grounded in local knowledge and address the real needs of the populace, thereby promoting sustainability and a sense of ownership.
- **Cultural sensitivity:** The agenda acknowledges the value of cultural sensitivity and makes suggestions to incorporate customs and beliefs into mental health practices in order to enhance their acceptability and efficacy within the context of climate change.
- **Evidence-based practices:** The agenda prioritises evidence-based practices, promoting research to understand the specific mental health impacts of climate change in the region and using this knowledge to inform interventions.
- **Capacity building:** The agenda focuses on enhancing the ability of community leaders and healthcare providers in SSA to recognise and manage mental health issues associated with climate change.

Limitations

- **Resource constraints:** Financial and time constraints impeded the mobilisation of the project to all countries in SSA. Lack of internet connectivity and sufficient human resources also stalled project deliverables at certain points in time. However, the SSA and global team worked around these issues to ensure project deliverable compliance. More broadly in relation to the SSA region, inadequate financial and human resources could prevent suggested interventions from being fully implemented. Investments in mental health within the context of climate change will need to be substantial.
- **Data gaps:** The lack of information regarding how climate change affects mental health in SSA could impede the creation of focused and efficient interventions for the research and action agenda. It is imperative to close these data gaps in order to make evidence-based decisions.

Next steps

Taking this research and action agenda forward involves multiple stakeholders, including funding agencies, governments, civil society, educators and researchers. Below are suggestion next steps for the region:

- **Resource mobilisation:** Securing financing from governments, donors, corporations and international organisations to undertake research on the priority themes identified in this agenda and translate this knowledge into action.
- **Data collection and analysis:** Investment in extensive data collection initiatives to close the knowledge gaps regarding the effects of climate change on mental health in SSA. This should include longitudinal studies to provide insights into evolving trends and inform ongoing intervention strategies.
- **Policy advocacy:** Using this agenda and its outputs to advocate for the integration of mental health into climate change policies. Strides must be made together with legislators in order to ensure that national and regional strategies prioritise mental health across the region.
- **Collaboration and networking:** Building on the collaborative network built throughout the CCM project. The exchanging of research results, strategies and first-hand experiences must continue to be fostered in order to advance regional and global knowledge of mental health and climate change.

Conclusion

The urgency of addressing climate change and its impact on mental health in SSA cannot be overstated. As the region experiences more frequent and severe climate-related events – from extreme weather to shifting disease patterns – it is critical to move quickly to lessen the effects and strengthen community resilience.

Our comprehensive action and research agenda has proposed a multimodal strategy for addressing the relationship between mental health and climate change in SSA. Important tactics include using a trans- and interdisciplinary approach and focusing studies on the psychological effects and impacts of catastrophic weather occurrences, job loss, food insecurity, climate migration and environmental deterioration. Initiatives to improve resilience at the individual, community and institutional levels are essential. These may entail creating psychological support networks, encouraging adaptive coping strategies which include traditional healing methods and incorporating mental health concerns into efforts to adapt to and mitigate the effects of climate change at both the grassroots and policy level.

In order to address climate-related mental health issues and incorporate related interventions into current healthcare systems, it is imperative that mental health professionals and lay counsellors in SSA be given more training and support. Giving communities the power to engage in decision-making processes around mental health support and climate change adaptation will also ensure the creation of locally-relevant solutions. It is imperative that policymakers in the region prioritise incorporating mental health concerns into climate change policies and tactics. This could entail funding mental health services, supporting awareness campaigns and mainstreaming mental health across numerous sectors.

To conclude, addressing the intersection of climate change and mental health in SSA requires a multifaceted approach which integrates research, policy, cultural sensitivity and inclusiveness, community engagement, and international cooperation and funding. By prioritising this issue and implementing evidence-based action, we can mitigate the negative mental health impacts of climate change and build more resilient communities in the region.

Hearing from the Regional Community of Practice

“Well, for disaster management agency, they are the ones that give the response whenever any disaster issue is occurring. Either flooding, fire incident or [whatever]. So, the government is the head of the institution, so I think they are able to amend some certain situations they can handle.” (Participant, Dialogue 2, Breakout session 2)

“I would like to see a comparison between what has been the mental health state of the population living in normal weather conditions and what has been the mental state of populations living in areas affected by extreme weather conditions.” (Participant, Dialogue 2, Breakout session 2)

“You'll need to do research with the community, sharing into their problems and then try and find solutions for the community rather than what the researcher thinks is the problem and taking it to the community to investigate.” (Participant, Dialogue 2, Breakout session 2)

“I think it's important that we're calling the youth to help us to talk more on climate change, and also to sensitise people in their communities on climate change. And there are various ways that youth can come in.” (Participant, Dialogue 2, Breakout session 1)

“It's obviously a tricky subject in sub-Saharan Africa, as we all deal with very dysfunctional governments, but indeed, that is a resource, and they can be brought in in many ways.” (Participant, Dialogue 2, Breakout session 1)

“It's an interlinked [interdependency] between the human, animal, and environment, and it is a growing understanding or opportunity for research on the impacts of climate change.” (Participant, Dialogue 2, Breakout session 1)

“Much of what happens in the forms of research is...what I will call a more extractivist type of research, because it's mostly the Global North comes and do research, and then takes it back. But as a country, having something intentionally, structures that are set up, and those structures are going in and engaging in research, and basically translating that research through extensions to communities and creating education around it.” (Participant, Dialogue 2, Breakout session 2)

“...there's a much deeper backdrop of funding, and the fact that the academic canon is drawn from like five countries, and it's often drawn from white men. And so, I think there's a big narrative and norm, a dominant social norm, that we're needing to tackle.” (Participant, Dialogue 2, Breakout session 2)

“I'm thinking of the One Health... the One Health where we integrate the planetary health, the planetary health, of course, has to do with all the aspects of ecosystems, including issues of climate change, animals, including people, and even plants, so that we have it is an action into designing research.” (Participant, Dialogue 2, Breakout session 2)

“...one is building climate resilience through ecosystem-based adaptation planning.” (Participant, Dialogue 2, Breakout session 1)

“So, once a person's got depression, once they've got anxiety, once they've got that, then if you're lucky, there's a response. It's normally bloody poor. So, in a country where there are only ten psychiatrists, and I'm the psychologist, you're very dependent on lay counsellors.” (Participant, Dialogue 2, Breakout session 1)

“I do think what is the likelihood that we can be able to build such resilience without also instilling some fear in people? So, with regards to that, I'm not necessarily sure how that would work. But I would be really more

interested in ways that we can maybe suggest ideas on how to prevent the actual climate crisis themselves.” (Participant, Dialogue 2, Breakout session 1)

“I will have more conversations with more people to educate them about the importance of climate change and mental health in Africa.” (Chat, Dialogue 2)

“I will find and seek to partner with like minded individuals to actualize research into MH-climate change.” (Chat, Dialogue 2)

“As a journalist, I will focus more on climate change reports and how it affects mental health.” (Chat, Dialogue 2)

Glossary of terms

For a glossary describing relevant concepts and key words for the Connecting Climate Minds research and action agendas, please download from [here](#).

Who produced this report

Author contribution statement

Monika dos Santos drafted the manuscript, acted as a regional co-convenor, recruited expert participants, facilitated in the two dialogues, sourced local research ethical approval and undertook data analytics and research. Kenneth Yongabi Anchang sourced funding for the project, acted as a regional convenor, undertook project management, recruited expert participants, conducted interview and data analytics, and drafted content for the report. Emma Lawrance sourced funding for the global project and acts as a principal investigator, facilitated a dialogue, recruited expert participants, provided project oversight and proofread the final document. Nomfundo Kwini acted as a dialogue note-taker and facilitator, undertook transcript analytics and proofread the document. Iji John Eluwa acted as a dialogue presenter, project manager and proofread the final report. Manasi Kumar acted as a dialogue facilitator and proofread the report. Gustaaf Wolvaardt assisted with local research ethical approval, participated in the dialogues as an expert and proofread the report. Didi Thompson proofread and edited the manuscript.

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Conflicts of interest

All authors declare that they hold no conflicts of interest.

Appendices

Appendix 1: Dialogue agendas

Dialogue 1

Time (min s)	What	How
0-20	Welcome and introductions	
20-35	Visioning Exercise	Visioning Exercise: "Headlines" on Climate Centre Good Games platform Participants share what they are hoping to get from being part of Connecting Climate Minds, and why they are invested in this topic <i>Imagine we are in 2030 and the CCM project has received an important prize for its achievements in bringing CC/MH together in your region and making change for climate and mental health together through research and action. You are interviewed by a journalist as you are celebrating winning the prize.</i> Create / vote on and review Headlines created on the Good Games Platform
35-50	Scene setting	Brief context to the links between climate change and mental health What are we hearing in the region that we can come together and respond to: sharing highlights of pre-scoping Presentation on Connecting Climate Minds and Q&A
50-95	Breakout session 1: Regional needs	Host sets scene for breakout discussion Share climate hazards overview for the region

		Breakout groups Facilitator to lead the group in a discussion on the mental health challenges arising from climate change in the region. Questions posed to participants were: • <i>From your perspective, what are the climate hazards that concern you the most in terms of mental health and wellbeing?</i> • <i>For each hazard, discuss the following:</i> ○ <i>Who is being impacted? Are there any groups more at risk?</i> ○ <i>What are the mental health and wellbeing outcomes associated with this hazard?</i> ○ <i>How does this hazard specifically affect mental health and wellbeing; what are the mechanism e.g. housing, livelihoods etc</i> ○ <i>Encourage people to look at different types of mechanisms e.g. social, biological, cultural, environmental</i> ○ <i>Climate adaptation/mitigation actions - what is being done, what can be done?</i> ○ <i>Actions that are being/could be taken to improve or promote mental health and wellbeing in this context</i> Brief plenary highlights
95-105	Break	
105-110	Sneak preview	The cartoon artist shares one first draft reflecting some of the content of the session so far. Host invites participants to share in the chat how the image resonates.
110-160	Breakout session 2: Evidence gaps and research priorities	Breakout groups [same participants, facilitator and note-taker as first breakout session] Looking back at the hazard table from Breakout session 1 as a starting point, participants are guided to start thinking about where more research is needed. Hosts invite 2 or 3 facilitators to share some brief reflections
160-170	Cartoon reflection on dialogue output	Participants individually review each cartoon, and share their reflections. This can range from: suggestions for changes to make it more accurate, to sharing how it relates to our reality, or any other insights.

170-180	Wrap-up and next steps	Host shares a high level recap of the session, thanks everyone and celebrates this first coming together of the regional community. Participants complete brief survey to provide feedback on the dialogue
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Dialogue 2

Time (mins)	What	How
0-5	World Map Exercise	Host asks participants to place a “star” icon where they currently are and then take the “heart” icon and place it where they would like to visit one day or a place they have been and really loved/feel especially connected to.
5-15	Welcome and introductions	
15-30	Sharing updates	Presentation on global and regional progress and outcomes of dialogue 1, including vision, priority research themes.
30-50	Priority Research Theme Feedback	Jamboard shared on screen. Host introduces priority research theme activity and asks people to consider the following questions: Research questions • <i>Is this a valuable theme? Would you add or change anything?</i> • <i>Share any ideas on datasets, metrics or methods that could be applied to help address this.</i> Research topics • <i>What would be most helpful to know about this topic to understand and respond to it?</i> • <i>What would you want to know first?</i> Music plays and participants work independently on Jamboard at their own speed.

50-55	Scene setting for breakouts	Host sets the scene for the breakout discussion. Interested in the 'how', which includes the principles and desired outcomes for research, policy and practice in the region. The discussion will enable targeting of action and investment to both implement the research agenda in the region, and to ensure evidence on the climate and mental health nexus is translated into policy and practice
55- 90	Break out 1	Eight breakout groups. The first 4 on the theme <i>“Creating knowledge through research”</i> (e.g. How can we best implement the research agenda?) and the other 4 on <i>“Fostering evidence-based policy and action”</i> (e.g. How can we translate the growing evidence base to action in policy and practice?). Each group uses a Jamboard to discuss: 1) What does this look like when it is done well? 2) What opportunities & enablers exist? 3) What challenges must we overcome? 4) Who should our partners and stakeholders be? Brief plenary highlights
90-100	Break	
100-105	Draw your feelings	Host welcomes everyone back and invites all who'd like to join to grab a piece of paper and a pen/marker (or digital) - and draw how they are feeling today. Anyone willing is invited to show their image - share some reflections.

105-125	Spectrum Mapping	Host introduces and explains the activity. Share screen of Jamboard. Designed to reveal the diversity of perspectives and options around different concepts and to organize them into a meaningful spectrum. This activity aims to illuminate the group's range of perspectives and to organize those perspectives into a continuum. Question: <i>What do these concepts mean to you?</i> Focusing on one board at a time, host asks everyone to silently generate a point-of-view around that concept and write it on a sticky note on the Jamboard. People can add more than one. Once the sticky notes are posted, the host works with the group to sort them into a horizontal range of ideas. Sticky notes that express similar perspectives or options should go next to each other. Sticky notes that seem to be outliers should stand alone; they may sometimes end up defining the limits of the range. Repeat with the remaining concepts. At the end, ask for observations and insights on the lay of the land. And if any perspective or option has been excluded. If so, add it and re-sort as necessary.
125 - 155	Break out 2	Host introduces the second round of breakout rooms and provides a 2-minute summary of the Jamboard contributions from the first breakout. Original 8 groups reconvene and expand and add to their initial thoughts. Groups are allowed to continue on the theme they were already discussing or switch to the other theme.
155- 175	I will ... I want...	Host asks everyone to personally reflect about how they will contribute to this space, and share an "I will" statement in the chat, but only press enter when told to do so. After a minute, host invites everyone to press enter at the same time to make statements appear. This is repeated with a statement about "Out of this community I would like to see..."

175	Wrap up and next steps	Host shares a high level recap of the session, thanks everyone and outlines next steps. Participants complete brief survey to provide feedback on the dialogue
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Appendix 2: Survey methodology

The pre-dialogue survey was intended to gather demographic information about the participants, their awareness of, and personal or professional experience with climate change and/or mental health issues. Accessibility requirements were also asked to ensure that dialogues could be designed to suit the needs of participants. The post-dialogue survey was intended to (a) collect additional feedback on the draft research topics after they had been reviewed on the basis of discussions held during Dialogue 2, and (b) collect feedback on the project as a whole. Both surveys were implemented on Qualtrics and the link was distributed via email.

The pre-dialogue survey was sent to dialogue participants only, while the post-dialogue survey (sent out after Dialogue 2) was sent to both dialogue attendees and the wider regional community, composed of individuals who had expressed interest in receiving project updates. Survey recipients were also asked to forward the post-dialogue survey to others within relevant networks. For those who had not attended the dialogue(s), the invitation to complete the survey included a Participant Information Sheet and a digital consent form. The dialogue participants had already consented as part of the dialogue process.

Appendix 3: Criteria for selection of priority research themes

Priority research theme selection criteria was generated based on discussion with experts globally across sectors, disciplines, and by drawing on selection criteria used in other research priority setting exercises.²³ Criteria was applied by consultation within the analysis team.

- Does the research theme answer an existing evidence gap?
- Does the research theme have potential to inform decision making in policy and practice?
- Does it reflect one or more of the following:
 - identified greatest mental health challenges linked to climate change in the region
 - identified greatest climate hazards in the region
 - evidence policymakers most need to address emerging needs / overcome barriers to action
 - alignment with what the experts by lived experience voiced as challenges in the region
- Could it be feasible to undertake this research?
- Is it framed to be understandable, specific and practical for the research community to meaningfully address? Could a researcher or a funder with relevant expertise understand this theme?
- Is it specific enough to be meaningful and guide focussed research (if too broad or vague it won't help target resources meaningfully)
- It helps funders invest in this area (consider what is regionally relevant here)

Appendix 4: Coding frameworks for the research agenda and action agenda

1. Coding framework for research agenda

Research Category	Sub-categories
1. Impacts, Risks and Vulnerable Groups This category is about improving our understanding of the ways in which mental health is affected by climate change. For example, what mental health outcomes are impacted or at risk, the prevalence, severity, economic and societal cost of these impacts, and who is more vulnerable to these impacts. This category also includes the ways we can go about getting this improved understanding of mental health impacts of climate change - the methods and metrics we need to assess and monitor mental health in ways that are relevant to climate change, contextually appropriate, comparable etc.	<u>Cross cutting considerations to keep in mind for all sub-categories:</u> Timeframe Geographical variation
	1.1. Research that focuses on the prevalence, severity and nature of the experience of different mental health outcomes/challenges/experiences affected by different aspects of climate change. This may include research to understand the emergence of climate-specific mental health experiences and their relationship to already defined mental health challenges.
	1.2. Quantifying the fraction of mental health burden (including mortality) caused by climate change.
	1.3. Understanding the risk factors to mental health that are caused or affected by climate change as well as protective factors.
	1.4. Identifying population sub-groups (e.g. demographics, livelihoods, life stage, pre-existing mental health challenges) who experience increased vulnerability to mental health challenges caused by climate change, and conversely those experiencing resilience to these effects (i.e. vulnerable groups).
	1.5. Quantifying the cost (e.g. economic, social) of the additional mental health burden caused by climate change and insufficient climate action.
	1.6. Methods research to identify the most appropriate ways to assess and monitor the mental health impacts of climate change [including adapting pre-existing scales, creating new ones, determining appropriate mental health metrics and indicators for inclusion in global processes like Lancet Countdown]. This can also include the need for cross-cultural validation and development of culturally appropriate metrics.

2. Pathways and mechanisms This category is about improving our understanding of <i>how</i> mental health is affected by climate change. We are interested in research themes that can help identify, categorise and understand the range of ways that climate change or climate action may act to affect mental health. This can include considering pathways and mechanisms that are biological, psychological, societal or environmental in nature, and may be direct or indirect. Note that mechanisms can include mechanisms to the development, maintenance, and/or resolution of mental challenges, so this includes also mechanisms relevant to guide development or understand workings of interventions	<u>Cross cutting considerations to keep in mind for all sub-categories:</u> <i>What factors are linked with increased vulnerability or increased resilience for the associated mental health outcomes.</i> 2.1. Categorising and understanding the societal mechanisms by which climate change negatively impacts mental health [e.g. changes to livelihoods, disruption to cultural practices, food and water insecurity, forced migration, political factors] 2.2. Categorising and understanding the environmental mechanisms by which climate change negatively impacts mental health [e.g. air pollution, reduced exposure to biodiversity] 2.3. Categorising and understanding the psychological mechanisms by which climate change negatively impacts mental health [e.g. how temperature affects cognitive changes relevant to symptoms of mental health challenges]. 2.4. Categorising and understanding the biological mechanisms by which climate change negatively impacts mental health [e.g. impacts of psychotropic medication on thermoregulation, neurodevelopmental factors]. 2.5. Understanding mechanisms whereby climate action or mental health interventions benefit climate and mental health (i.e. co-beneficial mechanism). 2.6. Methods research to identify the most appropriate ways to assess and monitor pathways and mechanisms by which climate change negatively impacts mental health and wellbeing (e.g. systems mapping across disciplines)
3. Mental health benefits of climate action [adaptation and mitigation]	3.1. Identifying climate actions that integrate or align with mental health benefits [co-beneficial climate actions, e.g. increased tree cover in cities] 3.2. Quantifying co-benefits of climate action for mental health (including number of people experiencing the benefit, size of effect, economic considerations).

This category is about how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health. This includes quantifying costs and benefits of climate action for mental health, understanding what is needed to support better alignment between climate action and mental health action, and identifying where this integration is already happening across strategies and policies.	3.3. Exploring how the mental health costs and benefits of climate action may differ across population sub-groups (e.g. demographics, livelihoods, life stage)
	3.4. Understanding the governance structures/decision support tools that enables alignment of action for climate change and for mental health across sectors
	3.5. Mapping and monitoring the integration of mental health within adaptation and mitigation policies across sectors [e.g. National Adaptation Plans, energy, transport, food, water, agriculture]
	3.6. Exploring opportunities for mental health to be integrated into other climate priority areas i.e. loss and damage and climate finance.
	3.7. Determining best approaches for climate action (e.g. emissions reductions or climate adaptation) within the mental health sector (ensuring psychiatric facilities can be kept cool in heat waves; green space projects in mental healthcare facilities)
	3.8. Methods research to identify the most appropriate ways to assess and monitor mental health benefits of climate action [e.g. place-based approaches, methods for attributing and quantifying co-benefits, methods for assessment of the mental health implications of decisions in other sectors]

4. Mental health interventions/solutions in the context of climate change This category is about identifying the most effective mental health interventions/solutions to support mental health in the context of climate change. This might be about providing support to people already experiencing negative mental health impacts, or about reducing risk or severity of future negative mental health impacts. This includes learning from knowledge held by different disciplines, communities and cultures, understanding how existing mental health interventions are affected by climate change, identifying and evaluating existing interventions that are relevant to the context of climate change, and developing new interventions. Interventions are relevant at all levels (individual, family, community, systems) and across sectors.	<u>Cross cutting considerations to keep in mind for all sub-categories:</u> LEVEL (e.g.) Individual, Family, Community, Systems MECHANISM (e.g.) Biological, Psychological, Social, Environmental SECTOR (e.g.) Education, Healthcare, Public Health Effectiveness considerations include impacts across different population groups, and implementation considerations might include providers, cost and time. 4.1. Understanding different ways of knowing, being and doing in different cultures and communities that can build individual, community and ecological resilience 4.2. Understanding how existing mental health interventions are affected by climate change 4.3. Identifying and evaluating mental health interventions that are already designed for or relevant to the context of climate change and/or integrate climate change considerations 4.4 Amending, implementing and evaluating relevant mental health interventions from other settings to be appropriate for climate-related impacts? 4.5. Co-designing, implementing and evaluating novel interventions that meet climate-related mental health needs
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	4.6. Comparing cost-effectiveness, implementation considerations, and effectiveness across interventions for a particular setting and particular population group to determine "best buys"
	4.7. Identifying, developing and evaluating approaches to awareness-raising and capacity building to upskill workforces to recognise and respond to the mental health impacts of the climate crisis (e.g. mental health professionals, emergency responders)

2. Coding framework for action agenda

Action Category	Sub-category
1. Creating an enabling environment for research at the intersection of climate change and mental health	1.1 Desired state of research - This code captures <i>what good looks like</i> for climate change and mental health research in the region that implements the research agenda. - What are the features of the kind of research that is desired or valuable? Are there specific attributes or milestones that would signify this state of research?
	1.2 Opportunities and enablers This code captures opportunities to progress the climate and mental health research field in the region towards the desired state, and factors that would enable progress. May be general or specific, and may link to what is required to overcome the challenges outlined in the next code.
	1.3 Challenges holding back research This code captures challenges that are stopping the climate and mental health field in the region from currently being in the desired state, or are predicted to emerge in trying to create investment in and implementation of the research agenda.
	1.4 Partners/stakeholders This code captures any key individuals, organisations or stakeholder types identified as being particularly important to engage for implementation of the research agenda in the region and securing required investment.
	1.5 Priority next steps/recommendations to investors and actors

	• This code captures the concrete next steps that need to be taken as priorities to create the conditions in the region to implement the research agenda. • This section will be used in the agenda to inform potential investors and key actors/decision makers where the priorities should be for next steps.
2. Translating a growing evidence base into action that can respond to the mental health impacts of climate change	2.1 Desired state of evidence to action in policy and practice • This code captures <i>what good looks like</i> for action on climate change and mental health in the region based on current and future evidence. • What are the features of the kind of pathways for translating evidence into action that are desired or valuable? Are there specific attributes or milestones that would signify that evidence-based action is occurring?
	2.2 Opportunities and enablers • This code captures opportunities to progress evidence-based action on climate and mental health in the region towards the desired state, and factors that would enable progress. May be general or specific, and may link to what is required to overcome the challenges outlined in the next code.
	2.3 Challenges holding back action • This code captures challenges that are stopping desired actions to protect mental health from the climate crisis in the region, or to enable co-beneficial climate action. The code may also include challenges that are predicted to emerge in trying to ensure that current and future evidence translates into change on the ground and at all levels of policy and practice.
	2.4 Partners/stakeholders • This code captures any key individuals, organisations or stakeholder types identified as being particularly important to engage for translation of evidence into relevant action and securing required investment.
	2.5 Priority next steps/recommendations to investors and actors • This code captures the concrete next steps that need to be taken as priorities to translate the emerging evidence base on climate and mental health into action in policy and practice. • This section will be used in the agenda to inform potential investors and key actors/decision makers where the priorities should be for next steps.

Appendix 5: Data collection and storage

Dialogues

Dialogues were conducted virtually on Zoom. Dialogues and breakout groups were recorded and transcribed by third party providers (Way with Words and Absolute Translations), and Zoom chat comments were saved. In dialogue 1, Word documents were used to capture notes from breakout discussions. In dialogue 2, Jamboard was used to capture notes and for participants to directly contribute comments.

Surveys

Survey distribution and data collection was carried out using the online platform Qualtrics. All survey data was collected by Imperial College London and anonymous data shared with SSA for analysis.

Data storage and sharing

Data was stored and managed by Imperial College London using a secure server. SSA was a Joint Data Controller for the data provided to this project for SSA and responsible for securely storing and sharing data with Imperial College London and with regional analyst teams. Data will be stored by Imperial College London for 10 years after study completion.

Appendix 6: Climate hazards in sub-Saharan Africa^{19, 24}

Observed climate-related hazards in Sub-Saharan Africa and projected changes

To guide informed discussions on the current and potential mental health consequences of current and future climate change in the region, it was vital to ensure a grounding in climate science. The Red Cross Red Crescent Climate Centre (RCCC) conducted a mapping of previous and projected climate hazards across the region to inform the dialogues and research agenda. The following is a summary of the climate-related hazards that sub-Saharan Africa has experienced over the last 30 years, and their projected increases to approximately 2030.

It is important to emphasise that the data gathered for the past and future periods differ not only in spatial resolution but are also based on different underlying principles. The former are through observations or documented occurrences such as those reported by national weather services and are aggregated at a country-scale. The latter are from climate modelling based studies and assessment reports published by the Intergovernmental Panel on Climate Change (IPCC), which are at a more granular spatial scale. For the same reason, the categories of climate-related hazards in the past and future periods may at times not be identical, but are generally comparable or related. For instance, flooding that is reported as a relevant climate hazard in the last 30 years here is not a direct output of global climate models (GCMs). Instead, the underlying meteorological variable (rainfall) from the simulations of GCMs are used for deriving indices that can be considered as a proxy for potential flooding in a future warming climate.

Sub-Saharan Africa

The sub-Saharan African region is one of the most vulnerable to climate change. The region has regularly experienced strong droughts, wildfires, floods, landslides, storms, and extreme temperatures in recent years according to the international disaster database EM-DAT ([EM-DAT - The international disaster database \(emdat.be\)](https://emdat.be)). Note that the number of extreme events recorded (e.g. wildfires) may be under-reported due in part to the missing records in the EM-DAT data for various reasons, such as poor or limited network of satellite and surface observations over the region, or limited reporting from national meteorological services. In addition, the spatial aggregation of the data to country-scales makes it difficult to map the occurrence of the past hazards to human settlements. The total number of people affected as reported in EM-DAT therefore may not always be truly representative of the frequency of occurrences of the underlying climate hazard. For instance, a country in Sub-Saharan Africa may have experienced frequent heatwaves, but if these events were in uninhabited regions, the total number of occurrences in the Figure may be an underreporting.

Previous Droughts

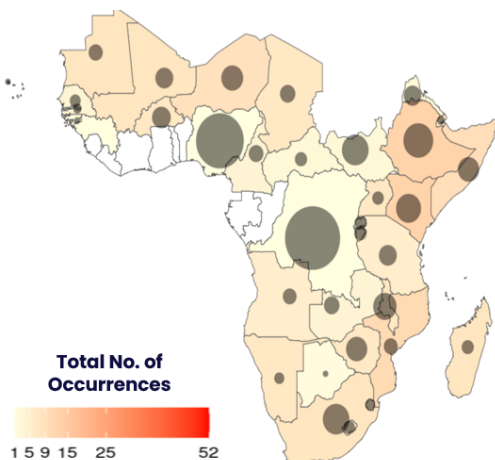
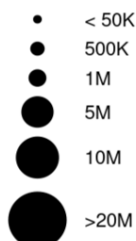
Droughts are the natural hazard that affected most people over the last 30 years, and strongly hit countries in the Horn of Africa affecting several million people. In the Democratic Republic of the Congo

(DRC) and Nigeria specifically, the EM-DAT lists only 1-2 droughts between 1991-2022, however the impacts affectedⁱⁱⁱ more than 20 million people on average. (Figure 1, top).

Drought and Wildfire

Env. Hazards Total Occurrences
& Avg. Affected Population:
1991 – 2022
(Source: EM-DAT)

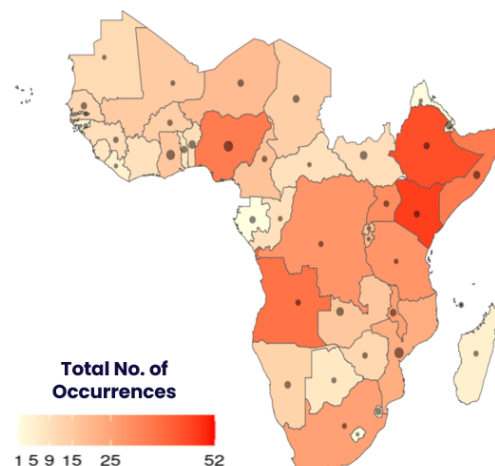
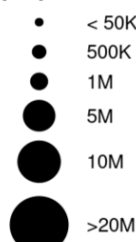
Average number of
people affected



Flood and Landslide

Env. Hazards Total Occurrences
& Avg. Affected Population:
1991 – 2022
(Source: EM-DAT)

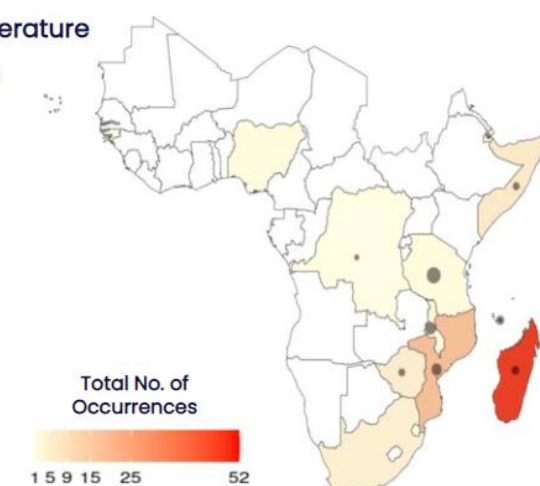
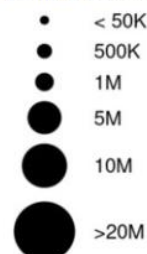
Average number of
people affected



Storm and Extreme Temperature

Env. Hazards Total Occurrences
& Avg. Affected Population:
1991 – 2022
(Source: EM-DAT)

Average number of
people affected



ⁱⁱⁱ Note, EM-DAT definition of “total affected” accounts for “the total of injured, affected, and homeless people”.

Figure 1. Number of reported droughts and wildfires (top), flood and landslides (middle) and storm and extreme temperatures (bottom) between 1991-2022. Red legend depicts the total number of occurrences of each category of hazard between 1991-2022, the black circle size is representative of the average affected population in that period.

Previous Floods

The number of experienced floods is also particularly high, reaching up to 50 floods in total between 1991-2022 in Ethiopia and Kenya (Figure 1, middle). Around 35 floods are recorded to have occurred in Angola and Nigeria affecting several thousand people.

Previous extreme (hot) temperatures and storms

Mozambique and Madagascar are the counties mostly affected by extreme (hot) temperature and storms (Figure 1, bottom). 19 and 52 such disasters are recorded to have occurred in Mozambique and Madagascar respectively, affecting less than 100,000 people.

Future projections

Future projections are based on the middle of the road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the CMIP6 multi-model GCMs ensemble provided in the IPCC, 2021. While not the worst-case emissions scenario, SSP2-4.5 assumes that the Paris agreement commitments are not achieved and 2°C global warming is not avoided. It is worth emphasizing that the low-level of confidence across GCMs when projecting certain climate hazards in the future (as indicated by the diagonal lines in the maps below in Figure 2) is not uncommon. GCMs have varying climate sensitivity and generally also differ in their representation of earth system processes. While their projections in the intensity of a particular hazard in the future may therefore differ despite a persistent signal in historical observed records, the direction of the signal (i.e., a positive or negative change of a climate hazard) are generally more robust across the GCMs.

- Mean and extreme temperatures: Future changes show a strong increase (high confidence). More than 40 additional hot days above 35°C are projected for 2030 compared to 1986-2005 for the Horn of Africa, western African Coast and southern Africa (high confidence) (Figure 2a).
- Droughts and wildfires: Droughts and risk to wildfire are projected to increase in the Southern Africa region, though there remains uncertainty across the range of GCMs (low to medium confidence) (Figure 2b).
- Heavy precipitation events and flooding: frequency and intensity of heavy precipitation events are projected to increase almost everywhere in Africa with additional global warming (high confidence, Figure 2c). This will be associated with a likely increase in pluvial flooding (high confidence).
- Sea-level rise: relative sea level rise is likely/virtually certain to continue around Africa, contributing to increases in the frequency and severity of coastal flooding in low-lying areas and to coastal erosion along most sandy coasts (high confidence).
- Tropical cyclones: models generally project an increase in the proportion of Category 4-5 tropical cyclones in Eastern Africa (low confidence), though the spread (indicating the range of change across models) is wide, with the median increase projected to be around 5% relative to 1986-2005.

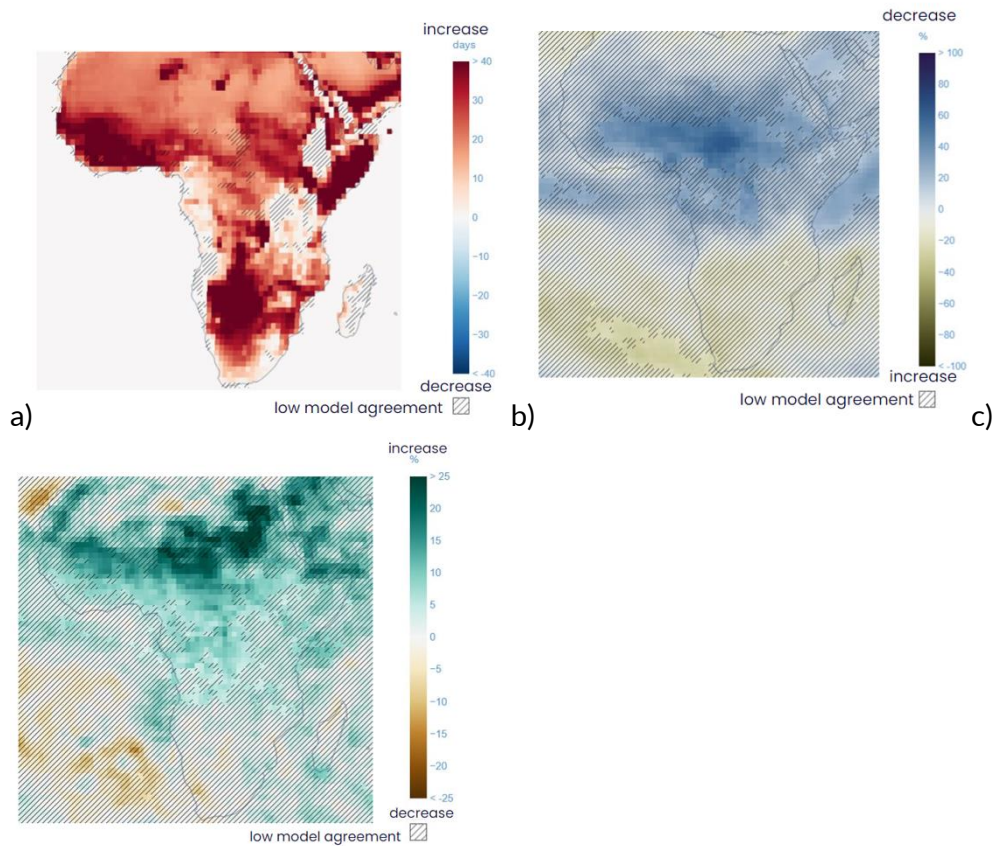


Figure 2. Projected changes in a) hot days above 35°C b) drought index SPI-6 and c) annual maximum 5-day precipitation for 2030 compared to 1986-2005 under the middle of the road emissions scenario SSP2-4.5.

Note: Detailed information [on the projections and the associated confidence](#) were accessed from the IPCC Working Group I contribution to the Sixth Assessment Report “Climate Change 2021: The Physical Science Basis” and the associated [IPCC WGI Interactive Atlas](#) and [regional factsheets](#).

Appendix 7: Summary of ten key regional needs

1. Global mental health in a changing climate: Understanding the sub-Saharan African experience

This theme aims to explore the unique challenges and magnitude of mental health issues arising from climate change in sub-Saharan Africa, emphasising the global significance of these concerns.

2. Breaking Stigmas, Building Resilience: Mental Health Interventions in the Context of Climate Change in sub-Saharan Africa

This theme delves into the prevalence of mental health stigma within sub-Saharan Africa and examines its impact on interventions, with a focus on strategies to break down stigma and enhance resilience in the face of climate-related challenges.

3. Indigenous Wisdom in Global Mental Health: Harnessing African Knowledge for Climate-Related Challenges

This theme investigates the efficacy of African Indigenous knowledge, traditional medicines, counselling techniques, and practices in promoting mental health, positioning these culturally rooted approaches as valuable resources in the global discourse on climate-related mental health challenges.

4. Syndemics and Trauma: Interconnected Challenges in Sub-Saharan Africa's Mental Health Landscape

This theme explores the complex relationships between wars, violence, trauma, epidemics, climate change-induced disasters, and their collective impact on mental health outcomes, emphasising the need for inter and trans disciplinary approaches to address these challenges.

5. Community Resilience: Coping Strategies and Adaptive Responses in the Face of Climate-Related Stressors

This theme explores the coping mechanisms and resilience strategies employed by communities in sub-Saharan Africa, highlighting the global significance of these local responses and their potential benefits for emotional wellbeing and mental health in the context of climate change.

6. Innovations in Mental Health Interventions: Adapting Global Strategies to Local Realities

This theme explores innovative approaches and interventions in the field of mental health tailored to the unique challenges posed by climate change in sub-Saharan Africa. It emphasises the importance of adapting global strategies to local contexts for effective and culturally sensitive mental health support.

7. Climate Change, Conflict, and Mental Health: Bridging Disciplines for Comprehensive Solutions

This theme investigates the intersection between climate change, conflicts, and mental health in sub-Saharan Africa. It encourages inter and trans disciplinary collaboration to understand and address the complex interplay of these factors, seeking comprehensive

solutions to promote mental health in regions affected by both environmental and societal challenges.

8. Technology and Telepsychiatry: Overcoming Barriers to Mental Health Access in sub-Saharan Africa

This theme focuses on the role of technology and telemedicine in overcoming geographical and infrastructural barriers to mental health access. It explores innovative ways to leverage digital solutions to provide mental health support and interventions in the context of climate change in sub-Saharan Africa.

9. Youth and Mental Health Resilience: Nurturing Future Generations Amid Climate Uncertainty

This theme centres on the mental health challenges faced by the youth in sub-Saharan Africa in the context of climate change. It explores strategies to nurture resilience among younger generations, acknowledging their unique vulnerabilities and strengths in the face of evolving environmental conditions, their unique vulnerabilities and strengths in the face of evolving environmental conditions.

10. Policy and Advocacy for Climate-Resilient Mental Health Systems: A Global Imperative

This theme addresses the crucial role of policy and advocacy in building climate-resilient mental health systems. It explores how international collaboration can influence policies to better integrate mental health considerations into climate change adaptation and mitigation strategies in sub-Saharan Africa and beyond.

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